(DEMO) From Natural Language to Live 5G API Calls: A CAPIF-Compliant, Multi-LLM Intent Orchestration Platform

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***Abstract*—We demonstrate an open-source platform that converts free-text operator intents into live CAMARA API calls. A two-phase large-language-model (LLM) planner selects the CAMARA service and builds a schema-grounded payload, and a three-layer deterministic validator (schema, telecom semantics, live CAPIF registry) gates every call. Quality of Service (QoS) requests are authorized by the PCF of a containerized Open5GS 5G core with UERANSIM-emulated UEs. A live request takes 2.2–4.8 s with GPT-4o-mini or Claude Haiku 4.5 (6.9–8.2 s with Gemini), of which LLM planning accounts for 78–99%. On 300 unique bilingual intents, the LLM planners reach an Intent-to-API Workflow Success Rate (IWSR) of 95.3–97.3%, compared with 83.3% for a keyword baseline. Code, testbed and data: https://github.com/doadurak/mini-telco-platform.**

***Index Terms*—CAPIF, CAMARA, intent-based networking, LLM orchestration, 5G API exposure, Open5GS**

I. Introduction

Zero-touch Beyond-5G management requires orchestration layers that translate high-level intents into standards-compliant API invocations [1]. 3GPP CAPIF [2] and CAMARA [3], [4] standardize how verticals discover and consume network capabilities, but closing the intent-to-API gap on a *live* exposure stack remains open. OpenCAPIF [5] provides exposure infrastructure without NL planning; Turkovic et al. [6] ran live CAMARA QoD with hand-written payloads; Maatouk et al. [7] identify hallucination as the key risk of telecom LLMs, and Liu et al. [8] report 40% JSON-schema compliance for ungrounded GPT-4. Our open platform combines live CAPIF onboarding, schema-grounded multi-LLM planning, a three-layer validator, and CAMARA execution down to a 3GPP PCF.

II. Platform Architecture

Planning: A FastAPI backend exposes `POST /orchestrate` (Fig. 1). Phase 1 selects one of four CAMARA services (QoD v1.1, QoS Profiles, QoS Provisioning, Location Retrieval v0.5) from a schema-grounded catalog; Phase 2 generates the payload from the operation's OpenAPI schema, which is injected into the prompt [9]. Intent-level QoS classes (e.g. `QOS_E_STREAMING`) are mapped by target bit rate to the operator profiles the network enforces (`QOS_E/S/M/L`).

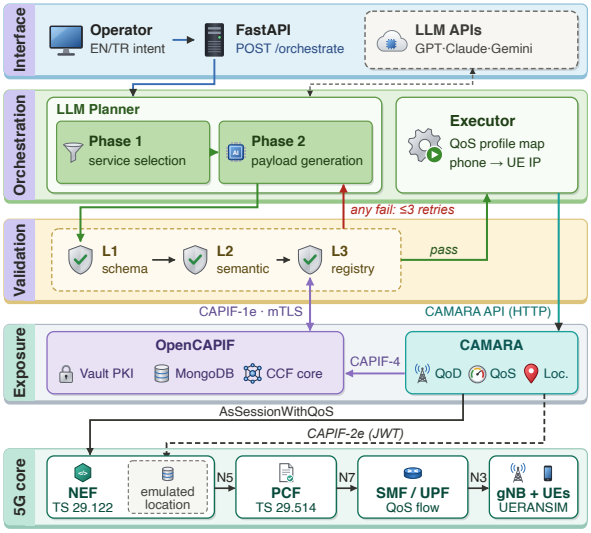


Fig. 1. Platform architecture. Any L1–L3 failure triggers at most three payload corrections; CAPIF-1e runs over mTLS, provider calls over HTTP.

Validation: L1 checks the payload against its Pydantic schema; the share of plans that pass L1 is the Schema Compliance Rate (SCR). L2 enforces telecom rules (E.164 numbers, QoS profiles that map to an operator profile, durations of 1–7200 s, unicast IPv4); the share that pass L2 is the Semantic Validity Rate (SVR). L3 queries the live CAPIF registry to confirm that the selected service is published. If the planner selects a registered but wrong service, L3 accepts it; such errors are only detected by comparison with the ground truth (IWSR).

Execution: For QoS, the adapter resolves the phone number to the UE's IP from the 5GC subscriber database; the CAMARA provider forwards an AsSessionWithQoS subscription to the NEF, which creates a PCF application session bound to the UE's PDU session. Location uses the NEF Monitoring Event API with emulated core data [10]. The CAMARA and NEF services are NCSRD's open-source implementations [11].

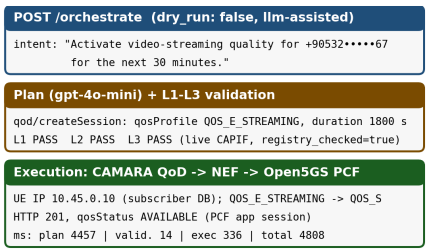


Fig. 2. Recorded live response (S1, GPT-4o-mini, `dry_run:false`, MSISDN masked).

TABLE I
300-ENTRY OFFLINE EVALUATION (ONE RUN PER INTENT)

Planner	IWSR	SCR	SVR	HR	SS
Keyword baseline	83.3%	100.0%	100.0%	16.7%	0.962
GPT-4o-mini	**97.3%**	100.0%	99.0%	**2.3%**	**0.975**
Claude Haiku 4.5	95.3%	99.3%	**99.3%**	4.0%	0.927
Gemini 2.5 Flash	96.0%	100.0%	98.3%	3.0%	0.945

III. Live Demonstration

The full stack of Fig. 1, including Open5GS [12] and UERANSIM [13], runs on one 16 GB host via `start.sh`. All scenarios run with `dry_run:false`, and their responses are in the repository. **S1 (QoD):** *"Activate video-streaming quality for ⟨MSISDN⟩ for the next 30 minutes"* yields `QOS_E_STREAMING`, 1800 s, enforced as `QOS_S`; the PCF accepts the session and QoD returns HTTP 201, `AVAILABLE` (Fig. 2). **S2 (Location):** the NEF returns a WGS-84 area (HTTP 200). **S3 (ambiguity):** dataset entry *"Provide network quality improvement for device ⟨MSISDN⟩ for a long time"* (ground truth: QoD). In the offline benchmark, GPT-4o-mini selected QoD, whereas Claude and Gemini selected QoS Provisioning; in the recorded live run, all three selected QoS Provisioning, which the PCF accepted (`ACTIVE`). Every plan passes L1–L3 and executes: the validator ensures that calls are well-formed and target registered services, not that the service matches the intent. **S4 (backend switch):** after each `POST /config/provider` switch, S1 yields identical CAMARA payloads from all three LLMs. UERANSIM lacks NGAP PDU Session Resource Modify, so the dedicated QoS flow is authorized by the core but not enforced on the emulated radio.

IV. Evaluation

The dataset holds 300 unique intents (152 EN / 148 TR; 96 easy, 94 medium, 110 hard): 275 across the four services and 25 ambiguous edge cases. We derived the ground truth (service, operation, payload checks) from the CAMARA OpenAPI constraints and NEF test fixtures; it supersedes a 60-entry pilot. IWSR is the share of plans that select the correct service and pass validation; HR is the share of wrong service selections. SS is one minus the standard deviation of IWSR across the three difficulty levels: it measures robustness to difficulty, not run-to-run repeatability. Planning uses temperature 0.1, so choices on ambiguous intents can vary between runs (cf. S3). The benchmark makes no provider calls; all 1,200 recorded plans pass the live L3 check. The keyword rules were written for the 60 pilot intents, on which the baseline scores 100%; on the 240 held-out intents it drops to 79.2%, while the LLMs reach 95.4–97.1%. The 8–14 failures per LLM are mostly confusions between QoD and QoS Provisioning (as in S3), and most occur on hard intents (IWSR 87–95%, versus 99–100% on easy and medium intents). *Fault injection* (Table II): each fault is caught by its target layer. Feedback repairs L1/L2 faults in one iteration but not an unpublished or wrong-but-registered service, because it only rewrites the payload. Without the schema in Phase 2, IWSR falls to 16.3%. *Scalability and external LLM APIs:* with live L3 checks, one backend process serves ≈160 req/s with the keyword planner; with GPT-4o-mini, latency stays at ≈2.6 s for up to 10 concurrent requests, so the LLM API is the bottleneck. If it fails, `auto` mode falls back to the keyword planner. As intents (with MSISDNs) reach an external API, production use needs pseudonymization or a self-hosted model.

TABLE II
FAULT INJECTION INTO CORRECT GPT-4O-MINI PLANS (LIVE L3)

Layer	Injected fault	Caught	Repaired	Time (iter.)
L1	missing field, wrong type	40/40	40/40	1.3 s (1)
L2	E.164, duration, profile, IPv4	80/80	79/80	1.3 s (1)
L3	service not published	20/20	0/20	3.8 s (3)

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